



Introduction to Numerai

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Overview

- Numerai is a data science competition where you build machine learning models to predict the stock market.
- Start with Numerai's free dataset made of clean and regularized financial data.
- The dataset is obfuscated so that it can be given out for free and modeled without any financial domain knowledge.
- Each row in the dataset corresponds to a stock at a specific point in time, represented by the era. The features are quantitative attributes (e.g P/E ratio) known about the stock at the time, and the target is a measure of stock market returns 20 days into the future.

id	era	feature1	...	feature310	target
n2b2e3dd163cb422	era1	0.75	...	0.00	0.25
n177021a571c94c8	era1	1.00	...	0.25	0.75
n7830fa4c0cd8466	era1	0.25	...	1.00	0.00
nc584a184cee941b	era1	0.25	...	0.00	1.00
nc5ab8667901946a	era1	0.75	...	0.25	0.25
n84e624e4714a7ca	era1	0.00	...	0.75	1.00

Modeling

- Your objective is to build machine learning models to predict the target.
- Here is an example model in Python using [LightGBM](#), but you can use any language or framework that you like.
- Full Official Example Model Link:
https://github.com/numerai/example-scripts/blob/master/example_model.ipynb

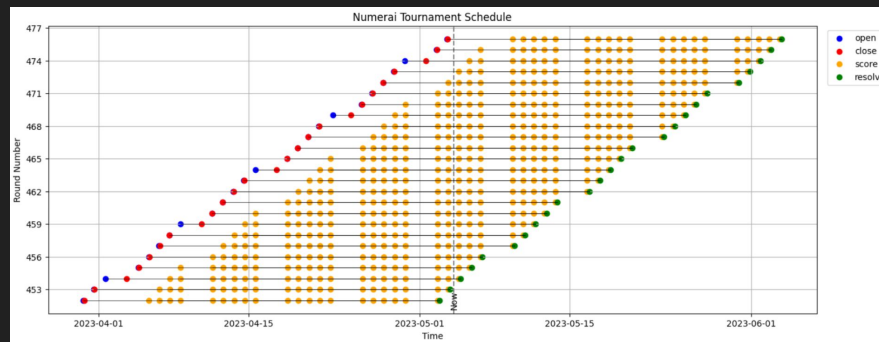
```
import lightgbm as lgb

model = lgb.LGBMRegressor(
    n_estimators=2000,
    learning_rate=0.01,
    max_depth=5,
    num_leaves=2 ** 5,
    colsample_bytree=0.1
)

model.fit(
    training_data[[f for f in training_data.columns if "feature" in f]],
    training_data["target"]
)
```

Submissions

- Every business day, new live features are released which represent the current state of the stock market. Your job is to generate live predictions and submit them to Numerai.
- The tournament is organized into rounds starting Saturday, Tuesday, Wednesday, Thursday and Friday every week.
- With each round spanning one month and 5 new rounds starting each week, we end up with an overlapping round schedule. At any given time, we can have up to 25 overlapping resolving rounds.
- Here is a visualization of the schedule of tournament rounds.



Submission Window

We generally expect rounds to follow this schedule:

Round	Open	Close
Sat-Mon	Sat 18:00 UTC	Mon 14:30 UTC
Tue	Tue 13:00 UTC	Tue 14:00 UTC
Wed	Wed 13:00 UTC	Wed 14:00 UTC
Thu	Thu 13:00 UTC	Thu 14:00 UTC
Fri	Fri 13:00 UTC	Fri 14:00 UTC

Actual open and close times may vary from round to round, but we will always maintain a minimum 1 hour submission window.

Submissions

To compete in the tournament you must submit live predictions in every round.

Here is an example of how to make a submission in Python using [NumerAPI](#):

```
# Authenticate
napi = numerapi.NumerAPI("api-public-id", "api-secret-key")

# Get current round
current_round = napi.get_current_round()

# Download latest live features
napi.download_dataset(f"v4.1/live_{current_round}.parquet")
live_data = pd.read_parquet(f"v4.1/live_{current_round}.parquet")
live_features = live_data[[f for f in live_data.columns if "feature" in f]]

# Generate live predictions
live_predictions = model.predict(live_features)

# Format submission
submission = pd.Series(live_predictions, index=live_features.index).to_frame("prediction")
submission.to_csv(f"prediction_{current_round}.csv")

# Upload submission
napi.upload_predictions(f"prediction_{current_round}.csv", model_id="your-model-id")
```

Submission Automation

Model Uploads

The simplest way to automate your submissions is to upload your entire prediction pipeline to Numerai. Once uploaded, Numerai will take care of running it every day to generate live submissions.

See model upload section for more details.

Official Cloud Solutions

If you prefer to run your own automation or have a more complex model or pipeline, we recommend an official cloud solution since we actively support these and any bug reports / support issues can be addressed by us directly. We have also engineered these to be highly reliable and very low cost.

- [Compute Lite](#) - use AWS and Sagemaker to deploy a medium complexity model
- [Compute Heavy](#) - use AWS or Azure to deploy a more complex pipeline

DIY for Google Cloud

If you're more comfortable with GCP, our users have provided some DIY solutions. These are not officially supported so you should already be comfortable with GCP if you choose one of these.

- [Google Cloud Function Example](#) - lower resources available, lower cost, simple
- [Google Cloud VM Example](#) - more resources available, variable cost, less simple

DIY for your local computer

If you prefer to keep things on-premises you can go with one of these. They are relatively simple and cheap, but running your own local computer comes with the burden of ensuring reliability and running your own tech support.

- [CRON example](#) - scheduling your script to run regularly and detect when to submit
- [NGROK example](#) - setup a webhook for Numerai to notify you when to submit

Late Submissions

- If you miss the submission window of the current round, you are still allowed to upload your submission but it will be considered "late".
- A late submission is still scored on CORR (and other secondary scores) but will not receive a TC score. This is because TC is a measure of your submission's contribution to the hedge fund returns and the meta model is constructed and trading happens immediately after the submission deadline. If your submission is late, it means that it was not included in the meta model and therefore has precisely 0 impact on the hedge fund returns.
- A late submission is also not eligible for staking and payouts for the round. The "at-risk" NMR number of a late submission is automatically set to 0 and excluded in the payout factor calculation.

Queued and Delayed Submissions

- If you miss the submission window of the current round, your submission will be automatically "queued" for the upcoming round.
- Queued submissions turn into on-time submissions immediately after the upcoming round opens. Since the ids of the live data changes every round, Numerai will automatically map your prediction's ids from the previous round to the latest round's ids.
- If your pipeline takes too long to run (ie over 24 hours), you may use your predictions generated for the previous round to submit in the current round. Again, since the ids of the live data changes every round, Numerai will automatically map your delayed prediction's ids from the previous round to the latest round's ids.

“Steaking”

When you are ready and confident in your model's performance, you may stake it with [NMR](#) - Numerai's cryptocurrency.

After the 20 days of scoring for each submission, models with positive scores are rewarded with more NMR, while those with negative scores have a portion of their staked NMR burned.

Behind the scenes, Numerai combines the predictions of all staked models into the stake-weighted Meta Model, which in turn is fed into the Numerai Hedge Fund for trading.

Staking serves two important functions:

- 1) "Skin in the game" allows Numerai to trust the quality of staked predictions.
- 2) Payouts and burns continuously improve the weights of the Meta Model.

How to stake

- Head over to numer.ai/staking, and click on the "manage stake" button to open the stake modal.
- Use the Stake form to stake NMR from your wallet on your model.
- Staked NMR will remain locked until you release it back to your wallet, which takes 1 month. Specifically, the NMR will be released at the resolution of your last active round at the time of the request. While pending release, the unstaked amount may still be subject to burns but will not count towards upcoming payouts.
- You can optionally configure where your earnings go and the score multipliers used to compute your payouts.

Stake

Release

Advanced

Stake Amount ⓘ

0.1NMR

Available: 0.450808483525612275 NMR (max) ⓘ

Confirm

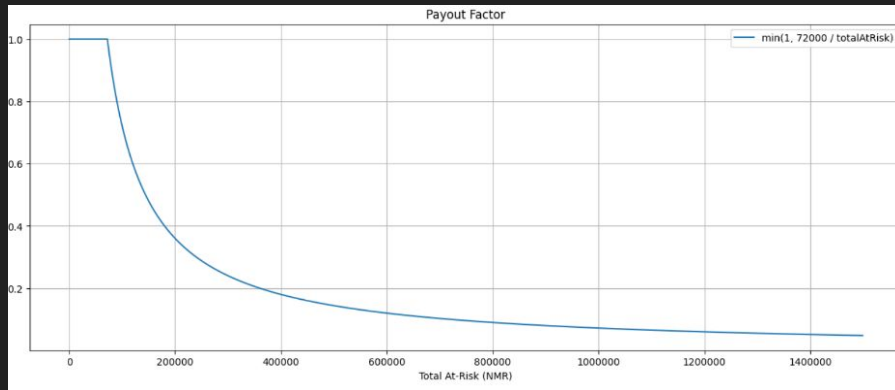
Payouts

Your payout is primarily a function of your scores. If you have a positive score you will get a payout. If you have a negative score a portion of your stake will burn.

The maximum payout or burn per round is capped at $\pm 5\%$

Stake is your model's stake value at the close of the round. This is also referred to as the stake value at-risk for a round. Your stake value at-risk for a round does not include any unstaked amounts that are pending release, and is set to 0 if you have no valid submission for a round.

`payout_factor` is a dynamic value that scales inversely with total NMR staked based on the `staking_threshold`.



`corr_mult` and `tc_mult` are configured by you to control your exposure to each score. Here are your options.

corr multiplier options	tc multiplier options
0x, 0.5x, 1.0x, 1.5x, 2.0x	0.0x, 0.5x, 1.0x

NMR

Numeraire (NMR) is the cryptocurrency that powers staking and payouts on Numerai.

To stake NMR on your model you must first deposit NMR into your Numerai wallet at your account's unique deposit address.

Available balance

0.440808483525612275 NMR

Deposit

Withdraw

Send NMR to

0x000000...000215ea 

Click to copy your deposit address

The deposit address only supports NMR. Anything else sent to it will be lost forever.

Get NMR

Scoring

There are two main scores used for payouts

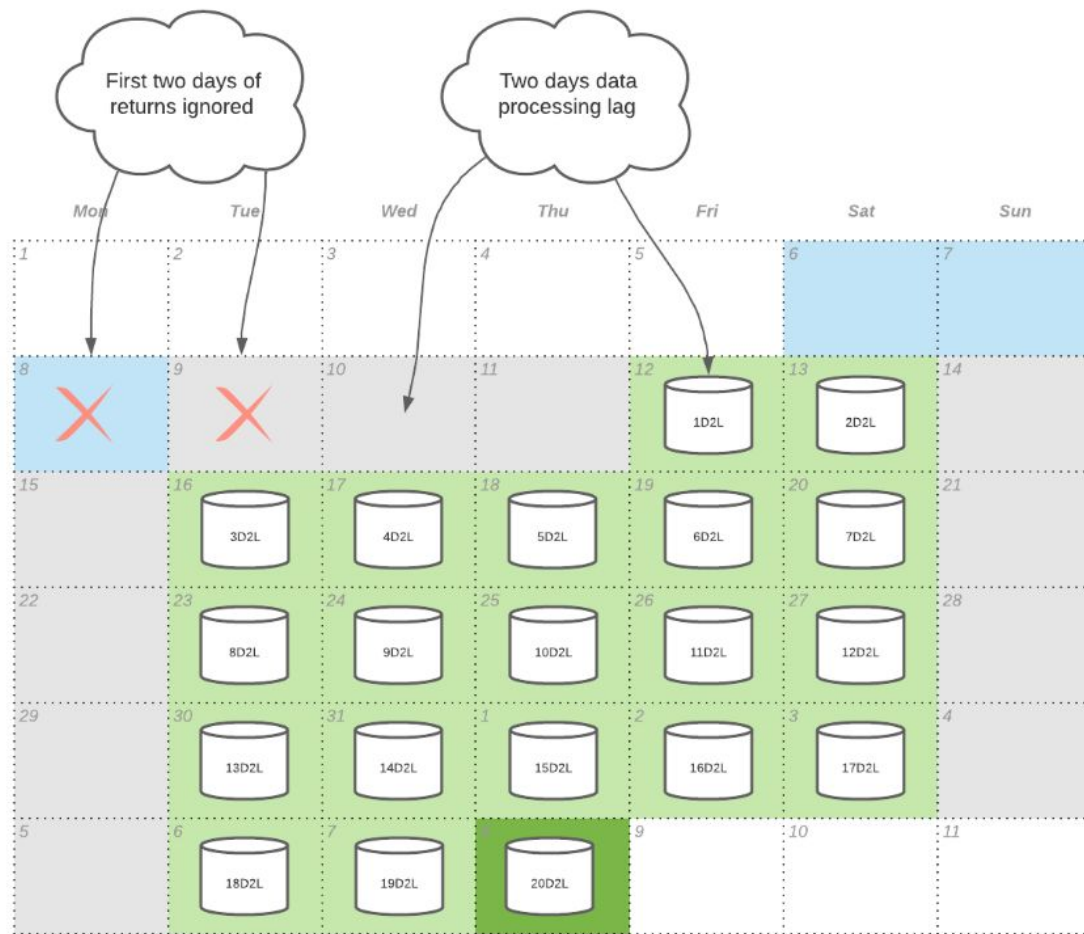
- [Correlation](#) (CORR): Your prediction's correlation to the target
- [True contribution](#) (TC): Your prediction's contribution to the hedge fund's returns

Numerai also has informational scores not used for payouts

- Feature Neutral Correlation (FNC): Your prediction's correlation to the target after neutralizing against all features
- Correlation with the Meta Model (CWMM): Your prediction's correlation to the Meta Model (stake weighted average of all predictions).

Scoring cont.

- Within a single round, submissions will receive 20 score updates until the final score of the round is computed.
- The reason why scoring is done over these 20 days is because the main target is built on 20 days of returns ignoring the first 2 days after round close. This is commonly referred to as "20D2L", where "20D" means "20 days of returns" and "2L" means "ignoring the first two days". Each score update is computed using an expanding window of returns.
- For example, the first day of scoring on Friday 12th uses a 1D2L target, which includes returns from Wednesday 10th only. The second day of scoring on Saturday 13th uses a 2D2L target which includes returns from Wednesday 10th through Thursday 11th. The final day of scoring 4 weeks later on Thursday 8th of next month uses a 20D2L target which includes returns starting from Wednesday 10th through Tuesday 6th of the next month.



The Leaderboard

- Only the final scores for rounds count towards a model's live performance.
- The 1 year average score is also called reputation and your model's rank on the leaderboard is primarily based on your model's 1 year average TC score.

STAKED MODELS

5,561

TOTAL NMR STAKED

1,055,844 NMR

AVG 1 YEAR RETURN

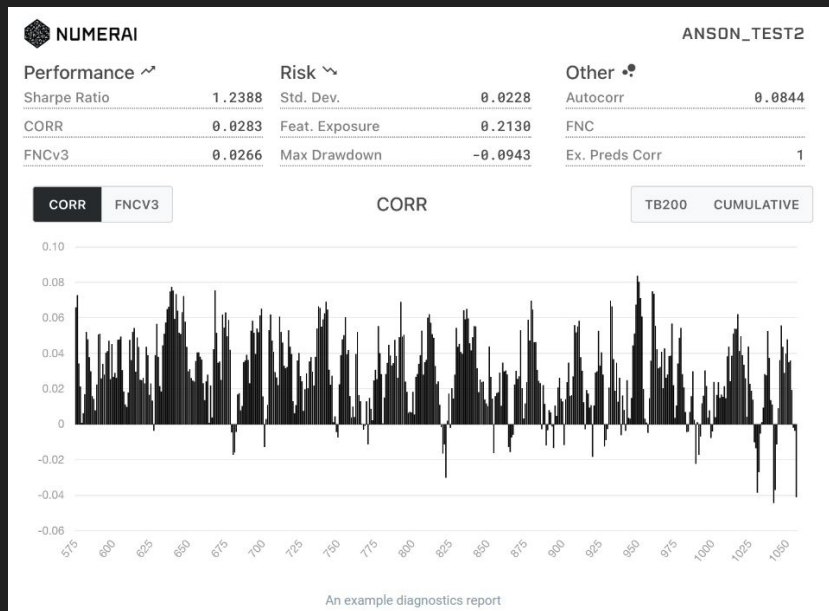
+62.07%

LEADERBOARD SEARCH		CORR20	CORR20V2	CORR60	FNCV3	↑ TC	STAKE	1 DAY	3 MONTH	1 YEAR
1	 SHATTEREDX SKTE	0.0246	0.0152	0.0419	0.0193	0.0365	23,497.24 NMR	3.2%	28.2%	615.9%
2	 K3_04 SKTE	0.0068	0.0071	0.0093	0.0091	0.0343	672.86 NMR	2.9%	—	—
3	 SHATT032 SKTE	0.0237	0.0127	0.0410	0.0186	0.0312	0.041 NMR	6.3%	22.7%	—
4	 EGG_19 SKTE	0.0189	0.0106	0.0383	0.0099	0.0309	0.027 NMR	4.8%	77.5%	—
5	 ML_IS_LYF SKTE	0.0305	0.0221	0.0474	0.0159	0.0302	0.15 NMR	2.5%	16.6%	63.9%
6	 EGG_2 SKTE	0.0138	0.0095	0.0258	0.0080	0.0300	97.97 NMR	5.9%	73.3%	—
7	 SANFRANCISCO SKTE	0.0182	0.0137	0.0342	0.0079	0.0281	0.070 NMR	3.9%	10.3%	49.9%
8	 HYDRAGIRAFFE SKTE	0.0210	0.0157	0.0336	0.0149	0.0280	26.3 NMR	9.2%	91.1%	133.5%
9	 SHATT021 SKTE	0.0264	0.0154	0.0413	0.0223	0.0269	0.041 NMR	0.40%	21.3%	—
10	 REPUTATION SKTE	0.0170	0.0095	0.0312	0.0069	0.0267	31,445.72 NMR	4.0%	62.7%	187.8%
11	 EGG_6 SKTE	0.0144	0.0086	0.0266	0.0061	0.0264	69.29 NMR	6.1%	28.9%	—
12	 EGG_20 SKTE	0.0188	0.0085	0.0357	0.0085	0.0263	0.021 NMR	5.9%	64.6%	—
13	 GOLDMN_6 SKTE	0.0202	0.0123	0.0401	0.0134	0.0263	—	—	—	—
14	 PRESIDIO SKTE	0.0202	0.0151	0.0398	0.0107	0.0260	1.6 NMR	4.3%	11.4%	52.5%
15	 SPOTTY_2 SKTE	0.0185	0.0118	0.0254	0.0124	0.0257	1.34 NMR	5.3%	58.8%	—

numer.ai/leaderboard

Diagnostics (Validation Scoring)

- Diagnostics is a tool to help you compute and visualize your scores over the validation dataset.
- If you uploaded your model via Model Upload, then Numerai will automatically run your model over the validation dataset to generate diagnostics.
- If you wish, you may also manually run diagnostics by heading over to numer.ai/scores and clicking on the Run Diagnostics button
- To note, all of the scoring code we use to generate diagnostics is also available in our [example scripts repository](#) if you wish to replicate this locally.
- A word of caution: past performance is no guarantee of future performance. This is especially true in the domain of financial machine learning. Take care not to rely too heavily on validation metrics during your research process to avoid overfitting to the validation dataset. If you train on the validation dataset, then don't expect your in-sample validation metrics to generalize out-of-sample.



What is CORR?

- The primary scoring metric on Numerai is the correlation between your model's predictions and the target.
- Numerai uses a special variation of correlation we call "Numerai Corr". At a high level, this metric is designed to be a good proxy for actual portfolio returns if the predictions were used in live trading.

First the predictions are gauss ranked. They do this to match our live trading process where all model predictions are standardized this way before being ensembled together in the meta model.

Then the target is centered around 0. They do this to match the gauss ranked predictions which are now centered around 0.

Finally both the gauss ranked predictions and the centered target are raised to the power of 1.5 before calculating the Pearson correlation. They do this to accentuate the tails as hedge fund tends to only trade the stocks with highest or lowest predicted returns.

What is TC?

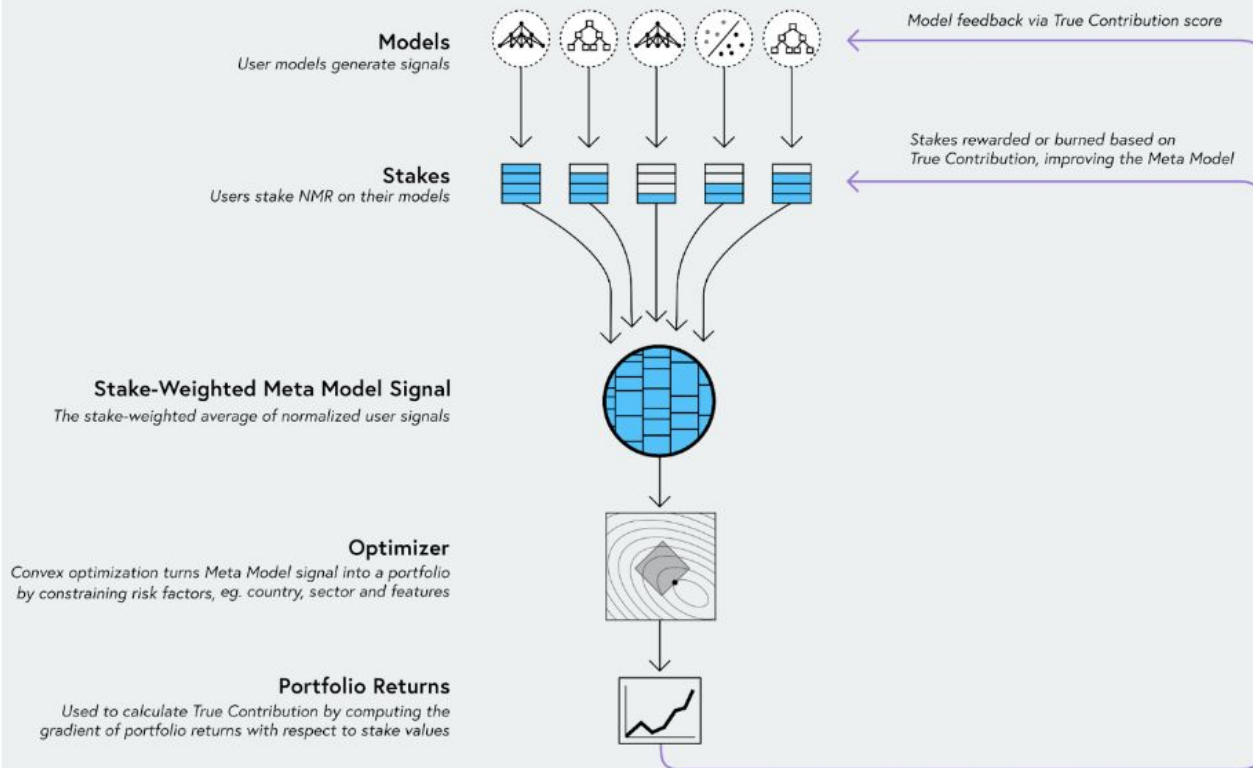
True Contribution is a measure of your model's contribution to the hedge fund's returns.

Read this [medium post](#) for the full background and motivation.

How is TC computed?

- Numerai combines submitted predictions via the stake-weighted average of every signal to create the Stake-Weighted Meta Model (SWMM). A data scientist who stakes a large amount of NMR on their model will have a larger weight in the Stake-Weighted Meta Model.
- The SWMM and hundreds of risk constraints (such as market, country and sector risk neutralization) are input into an optimizer. This transforms the SWMM into a realistic hypothetical portfolio.
- Numerai observes the hypothetical returns of that portfolio and calculates the gradient of the optimized portfolio return with respect to the stake. True Contribution is the magnitude of this gradient for each set of predictions constituting the SWMM.
- Numerai pays users based on a scaled version of this gradient magnitude. This is akin to Neural Network architecture and the idea of gradient descent.

TRUE CONTRIBUTION



How do I optimize for TC?

Numerai has found a few metrics seem to act as good indicators for TC. Currently, FNCv3 is the best metric we currently have. Numerai's Chief Data Scientist Michael Oliver also wrote a [forum post](#) discussing this topic more in detail

What is FNC?

Feature neutral correlation (FNC) is the correlation of a model with the target, after its predictions have been neutralized to Numerai's features.

Since features are known to be inconsistent on their own, models with too much linear exposure to features are expected to perform poorly. By neutralizing this linear exposure to features, FNC isolates the predictive performance of the model that isn't just from the feature exposure.

Calculation

To calculate a user's FNC for a given round we

- Normalize the predictions in their submission
- Neutralize their submission to Numerai's features for that round
- Calculate the Spearman rank-order correlation of their neutralized submission to the target

The current version of FNC shown on the website is called FNCv3 which is neutral to the "medium" subset of features in the V3 data.

THE DATA - V4.2(RAIN)

- Rain is the latest dataset release for Numerai. Rain contains 666 new features and 12 new targets, and entirely new feature groups to give Numerai data scientists a new way to do feature selection. Altogether, Rain contains 2132 features and 48 targets.
- Full Information:
<https://forum.numerai.ai/t/rain-data-release/6657>
- You can track the new example model's performance here:
numer.ai/lq_lgbm_v42_cyrus20



Thank You!

Numerai Github Example Scripts and Notebooks:

<https://github.com/numerai/example-scripts>

Twitter/Telegram: aventurine_eth

Email: aventurine.eth@proton.me

